1. **Your name and student ID**

   110062222 李子賢

2. **How to compile and execute your program and give an execution example**

README makefile

```
--How to Compile

    In "HW3/src/", enter the following command:
    $ make
    An executable file "hw3" will be generated in "HW3/bin/".

    If you want to remove it, please enter the following command:
    $ make clean

--How to Run

    Usage:
    $ ./hw3 <.txt file> <.out file> <dead space ratio>

    E.g., in "HW3/bin/", enter the following command:
    $ ./hw3 ../testcase/public1.txt ../output/public1.out 0.1
```

3. **The wirelength and the runtime of each testcase with the dead space ratios 0.15 and 0.1, respectively. Paste the screenshot of the result of running the HW3_grading.sh.**

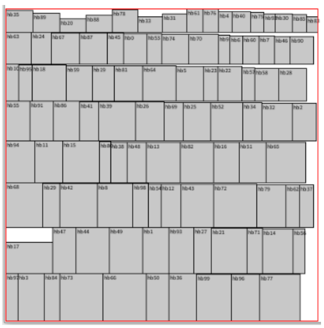
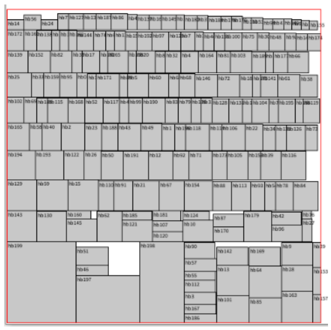
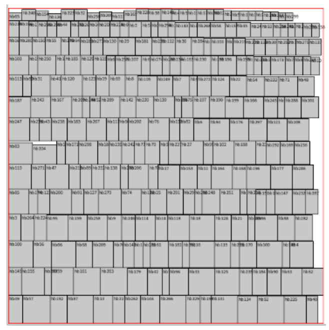
```
[u110062222@ic22 ~]$ cp CS6135_HW3_110062222.tar.gz HW3_grading/student/110062222/
[u110062222@ic22 ~]$ cd HW3_grading/
[u110062222@ic22 ~/HW3_grading]$ ./HW3_grading.sh
+-----------------------------------------------+
|                                               |
|    This script is used for PDA HW3 grading.   |
|                                               |
+-----------------------------------------------+
host name: ic22
compiler version: g++ (GCC) 9.3.0

grading on 110062222:
 checking item          | status
------------------------|--------
 correct tar.gz         | yes
 correct file structure | yes
 have README            | yes
 have Makefile          | yes
 correct make clean     | yes
 correct make           | yes

  testcase |       ratio | wirelength |     runtime | status
-----------|-------------|------------|-------------|--------
   public1 |        0.15 |     187877 |      158.20 | success
   public2 |        0.15 |     418573 |      590.07 | success
   public3 |        0.15 |     539213 |      590.50 | success
   public1 |         0.1 |     208259 |      160.53 | success
   public2 |         0.1 |     437676 |      590.30 | success
   public3 |         0.1 |     545651 |      592.56 | success
+-----------------------------------------------+
|                                               |
|   Successfully write grades to HW3_grade.csv  |
|                                               |
+-----------------------------------------------+
[u110062222@ic22 ~/HW3_grading]$
```

4. **Please show that how small the dead space ratio could be for your program to produce a legal result in 10 minutes.**

   這邊的 legal result 應該是指要能夠把所有 Block 放進去限制區域,沒有特別限制 Wirelength 要小於多少，又由於我的 Cost Function 跟面積無關，因此只需要看我的 Initial Floorplan 能不能成功就好。

Public1	Public2	Public3
		
0.0786	0.062	0.0557

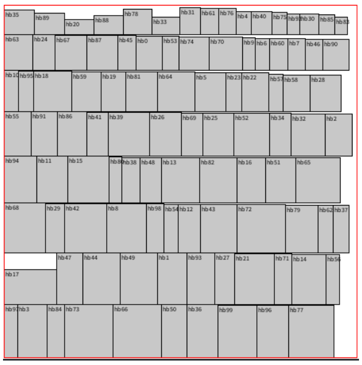
5. **The details of your implementation. If there is anything different between your implementation and the algorithm in the DAC-86 paper, please reveal the difference(s) and explain the reasons.**

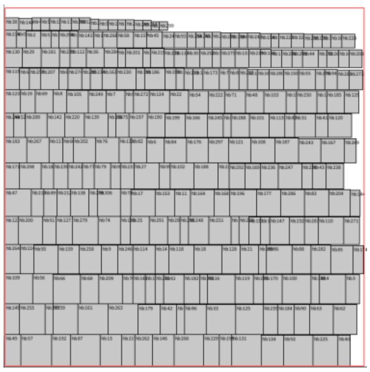
   DAC-86 Paper 中提到的 Initial Floorplan 是 12V3V4V…，但我測出來都沒辦法符合面積的限制。所以我會先套用我自己設計的 Initial Floorplan，把所有 Block 先放進限制區域中，在符合面積條件得情況下套用 SA 演算法不斷優化 Wirelength，如果 Perturb 的過程超出面積限制就不採用，並記錄 reject + 1，因此我的 Cost Function 只有比較 Wirelength 的長度，若 Initial Floorplan 沒辦法符合面積限制就過不了。

   另外我發現在 Perturb 的時候，M2、M3 能夠符合面積限制的機率比較低，M1 較容易發生，只保留 M1 的話會得到更好的結果。還有在做 M1 的時候，任何兩個 Operands 都可以交換，不限至於相鄰的 Operands。其餘部分皆與該 Paper 方法相似。

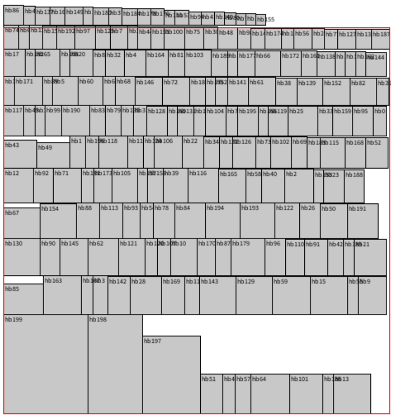
6. **Please describe your method of your initial floorplan.**

一開始我是按照每個 Block 的 max(height, width) 由大到小排進去 (可旋轉)，從左下開始一路放到右上，可以順利通過 Public1 和 Public3。

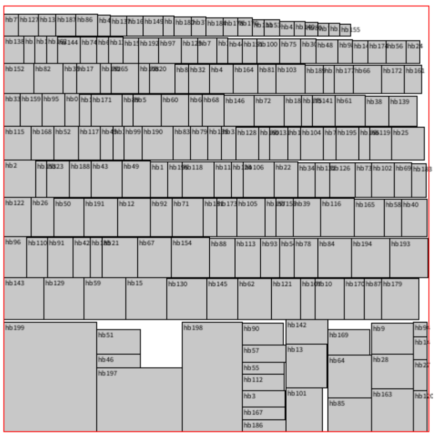

Public1


Public3

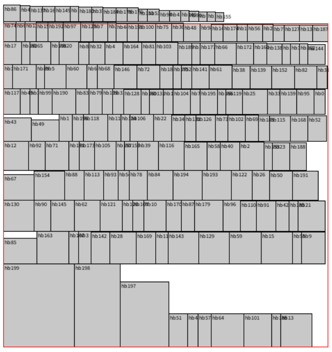
但是這樣沒辦法通過 Public2，因為有幾個特別大的 Block 會造成問題

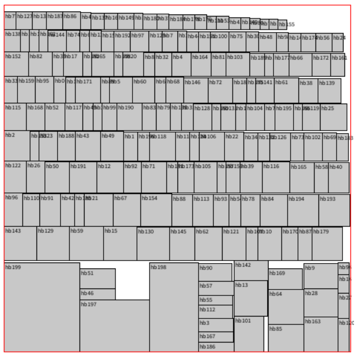


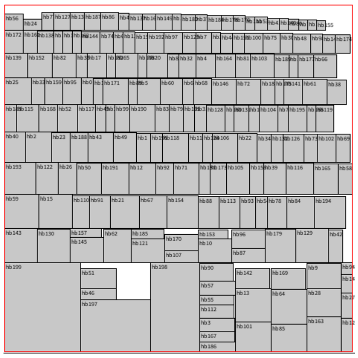
因此我採用在前幾列 **先放一個大的 Block，在其右邊再放進適當寬度的 Block**，上面幾列依樣按照原先的 max(height, width) 排進，這樣可以順利解決上述的問題

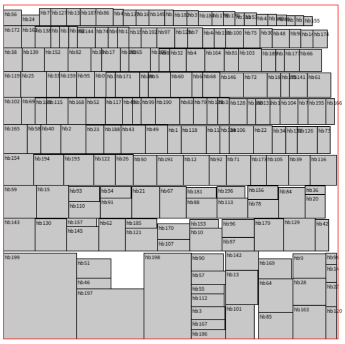


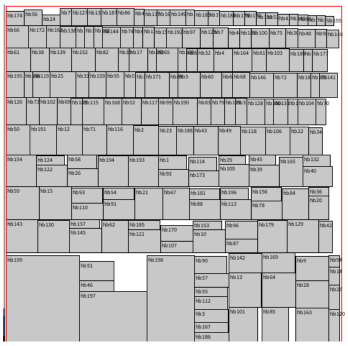
遵照這兩種列的排法，我會嘗試不同數量的組合，找出最合適的排法

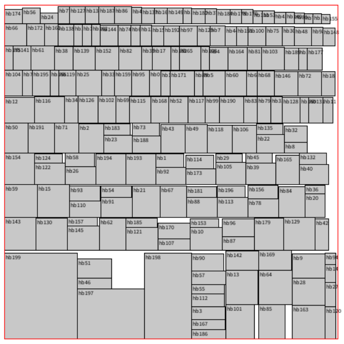












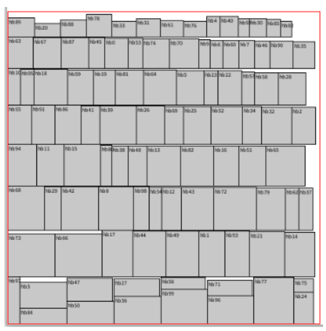
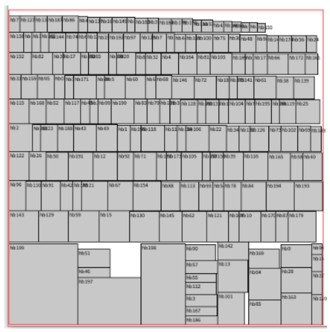
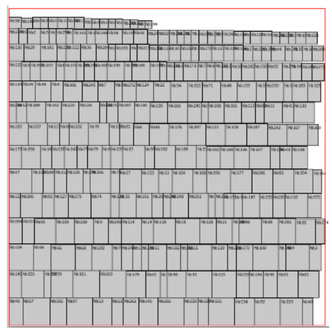
像是上面這樣產生多種組合，找出第一個符合面積限制的 Case 當作我的 Initial Floorplan

```
nthucad:~/HW3/src> ../bin/hw3 ../testcase/public2.txt ../output/public2.out 0.1
Target Area: 218406
cost: 27 wire_length: 603991
cost: 0 wire_length: 586834
cost: 0 wire_length: 579145
cost: 0 wire_length: 589362
cost: 0 wire_length: 593286
cost: 0 wire_length: 598086
cost: 0 wire_length: 591845
cost: 2 wire_length: 591933
cost: 3 wire_length: 593694
cost: 4 wire_length: 593202
0
Total Wire Length: 586834
nthucad:~/HW3/src>
```

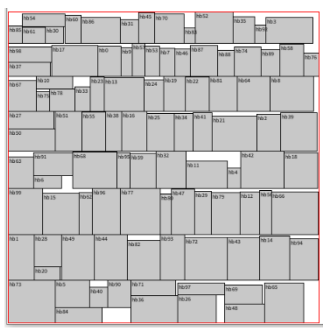
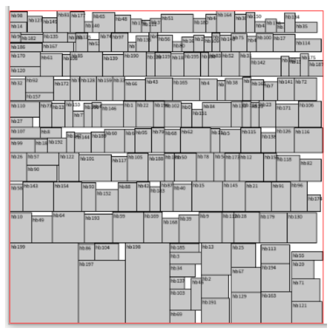
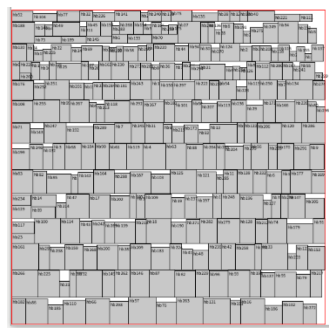
7. What tricks did you do to speed up your program or to enhance your solution quality?

先前提到，只套用 M1 的方法 (交換兩個 operands) 會讓 Solution Quality 更好，因此這邊比較 Initial Floorplan、套用 M1M2M3、僅套用 M1 三種 Case。

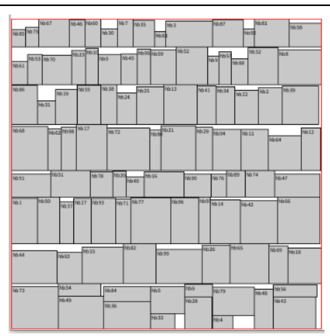
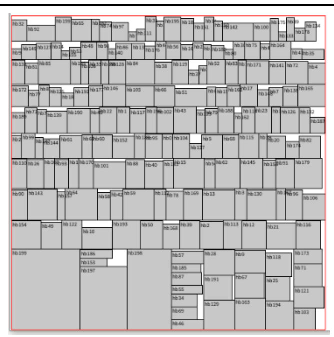
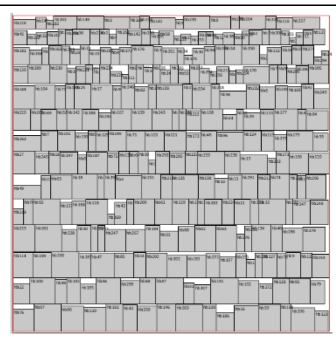
另外我有讓 SA 演算法盡量跑滿十分鐘，因此這邊不特別比較 Runtime，僅比較三種 Case 在 Dead Space Ratio = 0.1 下的 Wirelength 結果。

Public1	Public2	Public3
		
284739	586834	847185

Initial Floorplan

		
199720	427168	566083

套用 M1、M2、M3

		
194343	426793	551308

僅套用 M1

8. **What have you learned from this homework? What problem(s) have you encountered in this homework?**

我完全學到 SA 演算法的運作邏輯。遇到最大的問題是 Initial Floorplan 的部分，花了很多時間不斷修正參數和 Cost Function，都沒辦法藉由 SA 演算法來讓面積收斂。後來才決定讓 Cost Function 專注於減少 Wirelength，直接在 Initial Floorplan 階段處理完面積問題，花了很多時間設計不同的 Initial Floorplan 方法，還有測試不同參數的結果。